PLATFORM DEVELOPMENT SPECIFICATION

Skills Passport

Functional and technical requirements for building the Skills Passport platform

Version 1.0 · October 2026 · Scope: MVP and 12-month pilot

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1. Product overview

Skills Passport turns the real work of informally skilled young artisans (18–35) into a verified credential. After each job, the artisan sends photos and a voice note on WhatsApp. The client confirms the job with one tap, a master artisan reviews the quality, and AI maps the evidence to trade competency standards. The result is a shareable passport that buyers, employers, lenders and trade testing can trust.

Design principle: the passport must be useful from the first verified job. It must never depend on government recognition to deliver value.

Channels: WhatsApp for artisans; SMS and WhatsApp links for clients; mobile web for verifiers, field agents and buyers; public passport web pages with QR codes.

MVP scope

Module	MVP (pilot launch)	Later releases
Onboarding	WhatsApp sign-up, trade selection, consent, passport code	Self-service profile editing on web
Job logging	Photos, short video, voice note, client details; field-agent assisted mode	Offline capture app for field agents
Client confirmation	SMS and WhatsApp one-tap confirm, rating, reminders	USSD confirmation, voice-call confirmation
Verification	Verifier review queue with sampling rules, endorsements	Practical demonstration booking
AI	Transcription, job structuring, competency suggestions, image checks	Skill-gap voice lessons
Passport	Levels, public page, QR code, share link	Downloadable PDF portfolio
Buyers	Basic directory search and contact requests	Verified business accounts, subscriptions, hire tracking dashboards
Safety	Report and block a client	Trusted-contact location sharing and job check-in/check-out
Lenders	-	Consent-based verified work-history reports
Trade testing	Readiness indicator	Portfolio pack organised by unit standard

2. Glossary

Term	Meaning
Artisan	A young person with a practical trade skill, usually learned informally.
Job	One piece of completed work logged by an artisan, with evidence.
Evidence	Photos, video and a voice note describing the job, plus automatic time and location data.
Client	The person or business the job was done for. Clients confirm jobs; they do not need an account.
Verifier	An experienced master artisan registered to review job quality and endorse artisans.
Endorsement	A verifier's statement that the artisan's work meets the standard for their trade.
Competency unit	One skill in a trade's framework, e.g. "DC wiring and protection" for solar installation.
Passport level	Starter, Bronze, Silver or Gold, based on verified jobs, clients, endorsements and competencies.
Buyer	A business, NGO or household looking to hire an artisan.

3. Users and roles

Role	Access channel	Can do	Cannot do
Artisan	WhatsApp	Create passport, log jobs, see job status, choose which jobs and photos are public, share passport, accept or decline work requests, report a client, grant lender access	Confirm their own jobs or edit confirmed jobs
Client	SMS or WhatsApp link (no account)	Confirm or dispute a job, rate it, leave a comment, opt out of messages	See the artisan's other jobs or private data
Verifier	Mobile web	Review jobs in their trade, confirm competencies, rate quality, endorse artisans, request a practical demonstration	Review their own apprentices or relatives (declared conflict of interest)
Field agent	Mobile web	Enrol artisans, log jobs on an artisan's behalf (with the artisan present), help with photos and voice notes	Confirm jobs or endorse artisans
Buyer	Web	Search the directory, view public passports, send work requests, report the outcome	See artisan phone numbers before the artisan accepts
Lender (later)	Web	View a work-history report for an artisan who has granted consent	Access anyone without current consent
Admin	Web	Manage trades, competency frameworks, verifiers, field agents, fraud review, configuration, reports	Change job evidence

Authentication: artisans are identified by their WhatsApp number. Staff, verifiers and buyers sign in with phone or email, password and an OTP sent by SMS or authenticator app. Clients use single-use, signed confirmation links that expire after 7 days.

4. Core user journeys

4.1 Artisan onboarding

1. Artisan sends "Hi" to the Skills Passport WhatsApp number (or a field agent enrols them).
2. Choose language: English, chiShona or isiNdebele.
3. Age check (18–35) and residency confirmation.
4. Consent: what is collected, what is public, how clients are contacted. Voice-note version available.

5. Profile: first name and surname, trade(s) (up to 2), suburb or area, years of experience, how they learned (apprenticeship, family, self-taught, college).
6. Optional and separately consented: gender and disability, used only for inclusion reporting unless the artisan chooses a public badge (e.g. “Women in Trades”).
7. Optional profile photo.
8. System creates the passport code (format SP-{CITY}-{00000}, e.g. SP-HRE-00123) at level Starter and sends a short guide to logging the first job.

4.2 Log a job

1. Artisan chooses “Log a job” and picks the job type from a list for their trade (e.g. “Solar: battery and inverter installation”).
2. Send 1–3 “before” photos (optional for some job types).
3. Send 1–5 “after” photos and/or one video up to 60 seconds.
4. Send a voice note (up to 2 minutes) describing what was done, materials used and any problems solved. The bot prompts with three questions if the artisan prefers.
5. Enter client name and phone number (or share the client’s WhatsApp contact).
6. Confirm job date and area (defaults to today and the artisan’s area). Optionally share location.
7. Optional: amount charged (kept private by default).
8. Review summary and submit. The bot replies with the job reference and says the client will be asked to confirm.

After submission the system runs the media pipeline and AI processing (section 13), applies fraud checks (section 6) and sends the client confirmation request within one minute.

4.3 Client confirmation

```
SMS / WhatsApp to client:
"Skills Passport: Tendai says they did 'battery and inverter installation'
for you on 12 Nov. Was this work done? Reply 1 = Yes, 2 = No.
Rate the work: https://<short-domain>/c/8KQ2X  (Reply STOP to opt out)"
```

- Reply 1 or a tap on the link confirms the job. The link page shows the photos and asks for a 1–5 star rating and an optional comment or voice note.
- Reply 2 marks the job disputed. The artisan is told and the job goes to the review queue.
- Reminders at 48 hours and 5 days. No answer after 7 days: status client_expired. Field agents can follow up by phone and record the outcome.
- The client phone number must not belong to the artisan or any other registered artisan (section 6).

4.4 Verifier review

- Verifiers see a queue of jobs in their trade selected by the sampling rules (section 5), oldest first, with flagged jobs on top.
- Each item shows photos, video, transcript, AI summary and AI-suggested competency units with confidence.
- The verifier rates quality (meets standard, needs improvement, reject), confirms or removes each suggested competency, adds missing ones, and writes an optional note to the artisan.
- After reviewing at least 3 jobs by the same artisan rated “meets standard”, the verifier can endorse the artisan.
- A verifier can request a practical demonstration if evidence is unclear; the artisan is notified to arrange it.
- Target turnaround: 72 hours. Verifiers are paid per completed review.

4.5 Field-agent assisted logging

- For artisans without smartphones or data, a field agent logs the job on the agent's device in the artisan's presence.
- The job is marked agent_assisted with the agent's ID. The artisan receives an SMS with the job reference to confirm it is theirs.
- The same client confirmation and fraud rules apply.

4.6 Passport page

- Public URL (e.g. <short-domain>/p/SP-HRE-00123) and QR code. Server-rendered, under 200 KB, fast on 3G.
- Shows: first name and surname initial (artisan can show full name), trade(s), area at suburb level, level badge, number of verified jobs, average rating, number of distinct clients, competencies evidenced (verified vs. pending), selected job photos, endorsements, member since.
- Never shows: phone number, exact location, amounts charged, client details, disputes.
- Share options: WhatsApp share text, image card for WhatsApp Status, printable QR card.

4.7 Buyer search and work requests

1. Buyer searches by trade, area, level and minimum rating; results sorted by level, then rating, then recent activity.
2. Buyer opens a passport and sends a work request with job description, location area, preferred dates and contact details.
3. Artisan receives the request on WhatsApp and accepts or declines. On accept, both sides receive each other's contact details.
4. After 14 days, both sides are asked whether the job happened. A completed hire can be logged as a job in one tap, with the buyer as the client.

4.8 Lender report (later)

- Artisan chooses "Share with a lender" and selects a registered lender, or approves a lender's request.
- Consent lasts 30 days and can be revoked at any time. The artisan is notified each time the report is viewed.
- Report shows verified jobs per month, distinct clients, ratings, level history and client-confirmed amounts where available. Exported as a signed PDF with a verification code.

4.9 Trade-test readiness (later)

- When an artisan reaches Gold, the system generates a portfolio pack: evidence grouped by competency unit, verifier endorsements and a job list.
- Includes guidance on registering for the relevant trade test and any fee support available.

4.10 Job status transitions

From	To	Trigger
`draft`	`submitted`	Artisan confirms the job summary
`submitted`	`awaiting_client`	Media and AI processing done; confirmation sent to client
`awaiting_client`	`client_confirmed`	Client replies 1 or confirms on the link
`awaiting_client`	`client_disputed`	Client replies 2 or disputes on the link

From	To	Trigger
`awaiting_client`	`client_expired`	No answer after 7 days
`client_expired`	`client_confirmed or closed`	Field agent follow-up by phone
`client_confirmed`	`under_review`	Selected by sampling rules or flagged
`client_disputed`	`under_review`	Always
`under_review`	`verified, needs_improvement or rejected`	Verifier (or admin for fraud) decision

A job counts toward the passport when its status is client_confirmed, verified or needs_improvement, and never when it is rejected, disputed, expired or closed.

5. Verification and passport rules

Passport levels (defaults, configurable)

Level	Requirements	Shown as
Starter	Profile created	New member
Bronze	3 counted jobs from at least 3 different clients	Verified jobs
Silver	5 counted jobs, at least 4 distinct clients, at least 1 verifier endorsement, average rating 3.5 or higher, no open disputes	Endorsed artisan
Gold	12 counted jobs over at least 3 months, at least 8 distinct clients, endorsements from at least 2 different verifiers, at least 70% of the trade's competency units verified	Trade-test ready

- Levels are recalculated after every job status change. Levels can go down (e.g. an endorsement withdrawn or jobs rejected after a fraud review); the artisan is told why.
- Level history is stored for lender reports and analytics.

Review sampling

Rule	Default
First jobs of every artisan	First 3 counted jobs reviewed (100%)
Random sample after that	20% of counted jobs
Flagged jobs (fraud or quality signals)	100%
Disputed jobs	100%
Artisan with a dispute in the last 60 days	All new jobs reviewed
Before reaching Silver or Gold	At least 1 (Silver) or 3 (Gold) reviewed jobs rated "meets standard"

Competency status

- **Verified:** confirmed by a verifier on at least one job.
- **Pending:** suggested by AI with confidence of at least 0.80 on at least 2 client-confirmed jobs, not yet reviewed. Shown on the passport as "pending verification".
- **Gold** counts verified competencies only.

6. Fraud prevention

The credential is only worth something if buyers trust it. Every rule below creates a flag; flags send jobs to review and feed an internal trust score (0–100, never public).

Rule	Action
Client phone equals the artisan's phone, or any registered artisan's or verifier's phone	Block submission; ask for a different client
Same client phone confirms more than 2 jobs for the same artisan in 30 days	Flag; only 2 count toward the level
One client phone confirms jobs for more than 5 different artisans in 30 days	Flag all; possible confirmation ring
Duplicate or near-duplicate image (perceptual hash distance below threshold) across any jobs	Flag; exclude duplicate photos
Photo metadata shows capture date long before the job date (where metadata is available)	Soft flag
Image looks like a screenshot, stock photo or downloaded image (AI check)	Flag
Shared location far from the stated job area	Soft flag
Confirmation from a phone number registered in the last 24 hours on the platform	Soft flag
2 or more disputed or rejected jobs in 90 days	Freeze level changes; admin review
Verifier endorses an artisan they declared a conflict with, or approves at an unusually high rate	Flag verifier for admin review

Thresholds are configurable. Review flag rates monthly and tune to keep false positives low.

7. Competency framework

Each trade has a list of competency units. The framework is data, not code: admins manage trades and units through the dashboard, and each unit links to the matching national trade-test unit standard once that mapping is confirmed with the testing authority and master artisans.

Field	Description
`code`	Unique code, e.g. SOL-03
`title`	Short name shown to users and in all three languages
`description`	What a competent artisan can do
`evidence_guidance`	What photos and voice-note content prove it (shown to artisans and to the AI)
`job_types`	Which job types usually evidence this unit
`class_level`	Trade-test class the unit belongs to (to confirm with the testing authority)
`standard_ref`	Reference to the official unit standard

Pilot trades and draft units

Draft starting points only. Finalise each list with at least two master artisans per trade and align with official standards before launch.

Trade	Draft competency units
Solar PV installation	SOL-01 Site assessment and load calculation · SOL-02 Panel mounting and safe roof work · SOL-03 DC wiring and protection · SOL-04 Battery installation and safety · SOL-05 Inverter installation and set-up · SOL-06 Earthing and surge protection · SOL-07 Testing and commissioning · SOL-08 Customer handover and maintenance advice
Welding and metal fabrication	WLD-01 Reading drawings and measuring · WLD-02 Cutting and preparing metal · WLD-03 Arc welding, flat position · WLD-04 Arc welding, vertical and overhead · WLD-05 Fabricating gates, frames and burglar bars · WLD-06 Grinding, finishing and painting · WLD-07 Workshop safety and PPE
Plumbing	PLB-01 Pipe cutting and joining (PVC, copper, galvanised) · PLB-02 Water supply installation · PLB-03 Drainage and waste systems · PLB-04 Sanitary fittings (toilets, basins) · PLB-05 Geyser and solar water heater installation · PLB-06 Leak detection and repair · PLB-07 Pressure testing
Motor mechanics	MEC-01 Routine service · MEC-02 Brakes and suspension · MEC-03 Engine diagnosis · MEC-04 Engine repair and overhaul · MEC-05 Electrical systems and batteries · MEC-06 Transmission and clutch · MEC-07 Workshop safety
Tailoring	TLR-01 Taking measurements · TLR-02 Pattern drafting · TLR-03 Cutting fabric · TLR-04 Machine sewing and seams · TLR-05 Fitting and alterations · TLR-06 Finishing (hems, buttons, zips) · TLR-07 Garment design for customers

8. Conversation design

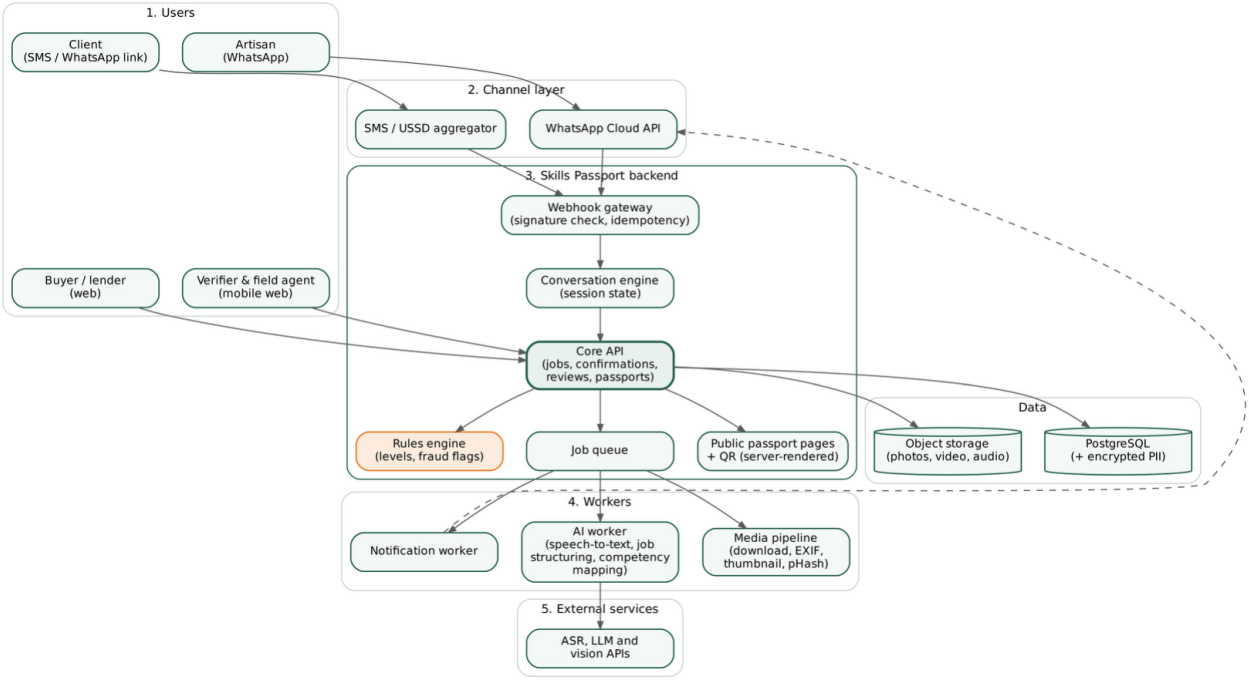
WhatsApp main menu (interactive list)

Option	Action
1. Log a job	Starts the job logging flow (section 4.2)
2. My passport	Level, verified jobs, rating, what is needed for the next level
3. Waiting for clients	Jobs awaiting confirmation; resend reminder; edit client number before confirmation
4. Share my passport	Link, QR image and WhatsApp Status card
5. Work requests	Open requests from buyers; accept or decline
6. Help	FAQ, talk to a field agent, report a client
7. Settings	Language, public profile options, lender access, delete my account

Messaging rules

- **Progress nudges:** after each confirmed job, tell the artisan exactly what is left for the next level (e.g. “2 more jobs from new clients to reach Bronze”).
- **WhatsApp templates:** needed for messages outside the 24-hour window: client confirmation (if sent by WhatsApp), work-request alerts, review results, reminders. Use the utility category.
- **Client messages:** short, clearly from Skills Passport, include opt-out. Send by SMS by default because clients may not have WhatsApp.
- **Media:** download WhatsApp media immediately after the webhook arrives, because media URLs are short-lived.
- **Guidance:** if a photo is too dark or blurry, ask for a retake straight away, with an example of a good photo for that job type.
- **Voice:** accept voice notes everywhere text is accepted; record key prompts in all three languages.

9. System architecture



Inbound messages flow down through the channel layer to the core API. Outbound messages (dashed line) leave through the notification worker.

Component	Responsibility
Webhook gateway	Receives WhatsApp and SMS callbacks; verifies signatures; deduplicates by message ID; queues processing
Conversation engine	Per-user state machine for onboarding, job logging and menus; session state in Redis with a TTL; resumes interrupted job logging
Core API	Business logic; REST API for verifier, field agent, buyer and admin dashboards
Rules engine	Passport level calculation, sampling selection, fraud rules; deterministic and unit-tested; thresholds stored as versioned configuration
Media pipeline	Downloads media, records metadata, strips metadata from public copies, creates thumbnails and compressed versions, computes perceptual hashes, checks quality
AI worker	Transcription, job structuring, competency suggestions, image checks (section 13)
Notification worker	Sends client confirmations and reminders by SMS or WhatsApp; artisan updates; delivery tracking
Public passport site	Server-rendered pages with caching; QR generation; Open Graph previews for WhatsApp sharing

10. Technology stack

Layer	Recommendation	Notes
Backend	TypeScript, Node.js 20+, NestJS (or Express)	Shared language with web apps
Database	PostgreSQL 16	Full-text search for the directory; add PostGIS later if location search needs it
Queues and cache	Redis with BullMQ	Media processing, AI jobs, reminders

Layer	Recommendation	Notes
Web apps	Next.js and Tailwind CSS	Public passport pages server-rendered; dashboards mobile-first
Object storage	S3-compatible bucket, private originals plus public compressed copies through a CDN	Signed URLs for private media
Image processing	sharp (resize, compress, metadata), a perceptual-hash library	Run in the media worker
WhatsApp	Meta WhatsApp Business Platform (Cloud API)	Verified business account and dedicated number
SMS	Zimbabwean SMS aggregator covering Econet, NetOne and Telecel	Two-way SMS needed for “Reply 1 or 2”
Speech-to-text	Evaluate Whisper (supports Shona) and specialist African-language ASR providers	Test isiNdebele and trade vocabulary on real samples
LLM and vision	Hosted multimodal LLM API with structured (JSON schema) output and a zero-data-retention agreement	Job structuring and competency suggestions
QR codes	Server-side QR generation library	Include passport code under the QR
Monitoring	Error tracking, uptime checks, structured logs	Alert on media pipeline and SMS delivery failures

11. Data model

Table	Key fields
`artisans`	id, passport_code, display_name, show_full_name, trades[], area_id, level, trust_score, status (active|frozen|deleted), language, experience_years, how_learned, gender (optional), disability (optional), public_badges[], joined_at
`artisan_pii`	artisan_id, full_name (enc), phone (enc), phone_hash, trusted_contact (enc, later)
`areas`	id, name (suburb), city, province, lat, lng
`trades`	id, code, name_en, name_sn, name_nd, active
`job_types`	id, trade_id, name, needs_before_photo, competency_unit_ids[]
`competency_units`	id, trade_id, code, title (per language), description, evidence_guidance, class_level, standard_ref, active
`jobs`	id, ref, artisan_id, trade_id, job_type_id, job_date, area_id, lat, lng, amount_charged, client_id, status, agent_assisted, agent_id, ai_summary (jsonb), public (bool), flags[], counted (bool), created_at
`job_media`	id, job_id, kind (photo_before|photo_after|video|voice), original_key, public_key, thumb_key, phash, captured_at, quality_score, public (bool)
`transcripts`	id, job_media_id, language, text, confidence, model_version
`clients`	id, phone_hash, phone (enc), name, confirmations_30d, reports_count, opted_out
`confirmations`	id, job_id, client_id, channel (sms|whatsapp|agent), token_hash, sent_at, reminders_sent, response (confirmed|disputed|expired), rating, comment, responded_at
`job_competencies`	job_id, unit_id, source (ai|verifier), confidence, status (suggested|confirmed|removed), reviewer_id
`reviews`	id, job_id, verifier_id, quality (meets|needs_improvement|reject), note_to_artisan, created_at
`verifiers`	id, name, phone (enc), trade_ids[], area_id, years_experience, conflicts (artisan_ids[]), status, approved_by

Table	Key fields
endorsements	id, artisan_id, verifier_id, trade_id, statement, created_at, withdrawn_at
buyers	id, org_name, type (business|ngo|household|government), verified, contact (enc), plan
work_requests	id, buyer_id, artisan_id, description, area_id, preferred_dates, status (sent|accepted|declined|expired), outcome (hired|not_hired|unknown), job_id
client_reports	id, artisan_id, client_id, reason, details, status
lender_consents (later)	id, artisan_id, lender_id, granted_at, expires_at, revoked_at; lender_access_log
level_history	id, artisan_id, from_level, to_level, reason, at
fraud_flags	id, entity (job|client|verifier|artisan), entity_id, rule, severity, status, resolved_by
configs	version, params (jsonb: levels, sampling, fraud thresholds), active_from
audit_log	id, actor_id, action, entity, entity_id, at, details

12. API endpoints

REST over HTTPS with JSON. Staff and buyers use JWT sessions; clients use signed one-time tokens.

Method and path	Purpose	Roles
POST /webhooks/whatsapp	Inbound artisan messages, media and status updates	Meta (signed)
POST /webhooks/sms	Inbound client replies and delivery reports	Aggregator
GET /c/{token}	Client confirmation page	Client (token)
POST /c/{token}	Submit confirmation, dispute, rating, comment	Client (token)
GET /p/{passportCode}	Public passport page	Public
GET /p/{passportCode}/qr.png	QR code image	Public
GET /reviews/queue?trade=	Verifier queue	Verifier
POST /jobs/{id}/review	Submit quality rating and competencies	Verifier
POST /artisans/{id}/endorse	Endorse an artisan	Verifier
POST /agent/artisans	Enrol an artisan	Field agent
POST /agent/jobs	Log a job on an artisan's behalf	Field agent
POST /media/upload-url	Signed upload URL (agent-assisted uploads)	Field agent
GET /directory?trade=&area=&level=&minRating=	Search artisans	Buyer, public (limited)
POST /work-requests	Send a work request	Buyer
POST /work-requests/{id}/outcome	Report whether the job happened	Buyer, artisan
GET /admin/flags?status=	Fraud review queue	Admin
POST /admin/flags/{id}/resolve	Clear or uphold a flag	Admin
CRUD /admin/trades, /admin/units, /admin/job-types	Manage competency frameworks	Admin
GET/PUT /admin/config	Levels, sampling, fraud thresholds (versioned)	Admin
GET /reports/pilot?from=&to=	Aggregate metrics	Admin

Example: AI job structuring output stored on the job

```
{
  "job_type": "solar_battery_inverter_install",
  "summary": "Installed a 3 kVA inverter and two 200 Ah batteries, wired DC isolator and breaker,
tested output.",
  "tasks": ["mounted inverter", "connected batteries in series", "installed DC isolator", "tested
system"],
  "materials": ["3 kVA inverter", "2 x 200 Ah gel batteries", "16 mm2 cable", "DC breaker"],
  "safety_measures": ["isolated supply before wiring"],
  "suggested_units": [
    { "code": "SOL-04", "confidence": 0.86, "evidence": ["photo_after_2", "transcript"] },
    { "code": "SOL-05", "confidence": 0.91, "evidence": ["photo_after_1", "transcript"] },
    { "code": "SOL-07", "confidence": 0.62, "evidence": ["transcript"] }
  ],
  "image_checks": { "quality_ok": true, "looks_like_trade_work": true,
"possible_stock_or_screenshot": false },
  "model_version": "llm-x.y / prompt v3"
}
```

13. AI components

Component	Input	Output	Guardrails
Speech-to-text	Voice note	Transcript, language, confidence	Low confidence: ask artisan to confirm a text summary
Job structuring	Transcript, job type, trade vocabulary	Summary, tasks, materials, safety measures (JSON schema)	Schema-validated; artisan sees and can correct the summary
Competency suggestion	Structured job, photos, unit evidence guidance	Units from that trade's framework only, with confidence and evidence references	Suggestions only; never verified without a verifier
Image checks	Photos	Quality score, trade-relevance, screenshot or stock-image likelihood	Flags only, never automatic rejection
Next-level coach (later)	Passport status, missing units	Plain-language tips and short voice lessons	Reviewed content library, not free generation

- Store model name, version and prompt version with every AI output.
- Build an evaluation set of at least 50 real jobs per trade, labelled by master artisans. Show AI suggestions to verifiers only once precision is at least 80% on that set.
- Send only job evidence and pseudonymous IDs to AI providers, never names or phone numbers.

14. Privacy, safety and compliance

Privacy

- Artisans control what is public: each job and photo has a public flag (default public for photos without people; private if a person or a house number is visible).
- Phone numbers are never public. Buyers get contact details only after the artisan accepts a request.
- Client phone numbers are used only for job confirmation, stored encrypted with a hash for matching, and clients can opt out at any time.
- Gender and disability are optional, separately consented, and used only for aggregate reporting unless the artisan opts into a public badge.
- Lender access only with explicit, time-limited, revocable consent; every view is logged and notified.

- Artisans can delete their account; public pages go offline immediately and personal data is deleted within 30 days (anonymised statistics kept).
- Strip location metadata from all public images; keep the original metadata only in private storage for fraud checks.

Artisan safety (especially young women working in clients' homes)

- MVP: report and block a client; a client with 2 or more reports is blocked from confirming jobs and from work requests.
- Later: before a job, the artisan can share the job location and expected finish time with a trusted contact. If they do not check out by then, the trusted contact gets an alert.
- Buyers must give a name and phone number; verified business accounts are marked.

Security controls

- TLS everywhere; role-based access enforced in the API; MFA for admin and verifier accounts.
- Signed, single-use, expiring client confirmation tokens; rate limits on confirmation pages and directory search to prevent scraping.
- Webhook signature verification and message-ID deduplication.
- Encrypted backups, audit log for admin actions and access to personal data, secrets in a vault, dependency scanning in CI.
- Security review before public launch of the directory.

Regulatory checklist

Requirement	Action
Cyber and Data Protection Act [Chapter 12:07]	Consent records, purpose limitation, data subject rights (access, correction, deletion)
Data controller licensing	Register/licence the operating entity with POTRAZ and appoint a Data Protection Officer
Cross-border data transfer	Confirm POTRAZ requirements before hosting outside Zimbabwe
Credit information (lender reports)	Before launching lender reports, check whether sharing work-history data with lenders falls under credit-reporting rules
Trade testing	Agree with the trade-testing authority how portfolios may be used; never claim official certification
Messaging	Opt-out on every client SMS; follow WhatsApp Business and SMS sender-ID rules

15. Non-functional requirements

Area	Requirement
Speed	Client confirmation sent within 1 min of submission; AI processing complete within 5 min; WhatsApp replies within 3 s
Public pages	Load in under 3 s on 3G; page weight under 200 KB excluding lazy-loaded photos
Media	Compress photos to about 200 KB for public display; video capped at 60 s; originals kept privately
Data costs	Artisans should spend less than 5 MB of data per logged job
Availability	99.5% monthly; queue and retry if WhatsApp, SMS or AI providers are down
Languages	Artisan and client content in English, chiShona and isiNdebele

Area	Requirement
Accessibility	Voice-first; field-agent mode for artisans without smartphones; screen-reader friendly web pages
Scale	MVP 500 artisans; designed for 50,000 artisans and 1 million jobs without redesign
Time zone and currency	Africa/Harare for display, UTC in storage; USD as integer cents

16. Analytics and reporting

Metric	Definition
Artisans enrolled	By trade, city, gender and disability (where consented)
Jobs logged	Total and per active artisan per month
Confirmation rate	Client-confirmed jobs / jobs sent for confirmation
Dispute and rejection rate	Disputed or rejected jobs / jobs logged
Time to level	Median days from joining to Bronze, Silver and Gold
Verifier turnaround	Median hours from job entering the queue to review
Flag rate	Flagged jobs / jobs logged, and share upheld after review
Work requests	Requests sent, accepted, and reported as hired
Income change	Self-reported monthly income at joining and at month 12 (short survey)
Trade tests	Artisans who sat and passed a trade test with a portfolio

Track product events (artisan_joined, job_submitted, client_confirmed, client_disputed, review_completed, level_changed, passport_shared, passport_viewed, work_request_sent, work_request_accepted, hire_reported) with pseudonymous IDs only.

17. Testing and quality assurance

- **Rules engine:** table-driven tests for every level rule, level downgrades, sampling selection and each fraud rule, including edge cases (the same client for two artisans, re-confirmation after expiry, deleted accounts).
- **Media pipeline:** test with real WhatsApp photos, videos and voice notes from low-end phones; duplicate detection on resized and cropped copies.
- **Conversation flows:** end-to-end scripts for onboarding, job logging (including interrupted sessions), confirmation and work requests in all three languages.
- **AI:** evaluation set per trade (section 13); re-run on every model or prompt change.
- **User acceptance:** test sessions with 20 artisans and 5 verifiers across the five trades before the soft launch.
- **Security:** test token expiry and reuse, scraping protection, and role permissions.

18. Build roadmap

Stage	Target date	Deliverables
Setup	Week 1	Repositories, environments, CI, Meta Business verification and WhatsApp number, SMS sandbox, database schema, draft competency lists for 5 trades

Stage	Target date	Deliverables
Pitch demo	By 3 Nov 2026	WhatsApp onboarding; log a job with photos and a voice note; AI summary and competency suggestions; client confirmation by SMS; public passport page with QR code. Test data only
MVP	End of pilot month 2	Verifier queue and endorsements, levels, fraud rules, field-agent mode, basic buyer directory and work requests, report a client, admin tools; soft launch with 30 artisans
v1.1	Months 4–7	Trusted-contact safety feature, verified buyer accounts, hire tracking, Bulawayo configuration, isiNdebele voice prompts
v1.2	Months 8–10	Lender consent reports, buyer subscriptions and payments
v2.0	Months 11–12	Trade-test portfolio pack, next-level coaching content, Pro upgrade for artisans

19. Open decisions

Decision	Options or next step
Official unit standards	Obtain trade-test unit standards for the 5 trades and confirm the responsible testing body
Level thresholds	Validate defaults with master artisans and 3–5 buyers
SMS aggregator	Two-way SMS, sender ID, cost per message, coverage
Hosting region	Depends on POTRAZ cross-border guidance
Speech-to-text vendor	Benchmark on real voice notes with trade vocabulary
Verifier payment	Rate per review and payout method
Public amounts	Keep amounts charged private (default) or allow artisans to show price ranges
Short domain	Register a short domain for passport and confirmation links
Buyer pricing	Per-hire fee vs. monthly subscription; test with pilot buyers